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A.P T.S KARNATAKA TAMILNADU MAHARASTRA DELHI RANCHI

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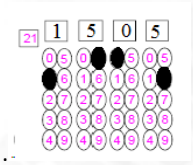
SEC: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **JEE-MAIN** **Date: 22-01-2026**

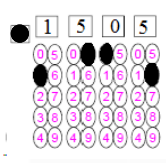
Time: 09.00Am to 12.00Pm **GTM-17** **Max. Marks: 300**

IMPORTANT INSTRUCTION:

1. Immediately fill in the Admission number on this page of the Test Booklet with **Blue/Black** Ball Point **Pen** only.
2. The candidates should not write their Admission Number anywhere (except in the specified space) on the Test Booklet/ Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **75 Questions.** The maximum marks are **300.**
5. There are **three** parts in the question paper 1,2,3 consisting of **Mathematics, Physics** and **Chemistry** having **25 Questions** in each subject and subject having **two sections.**
   (I) Section –I contains **20 Multiple Choice Questions** with only one correct option.
   Marking scheme: +4 for correct answer, **0** if not attempt and **-1** in all other cases.
   (II) Section-II contains **05 Numerical Value Type Questions.**
   - The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**

   To cancel any attempted question bubble on the question number box.
   For example: To cancel attempted Question 21. Bubble on 21 as shown below





Question Answered for Marking **Question Cancelled for Marking**

Marking scheme: +4 for correct answer, **0** if **not attempt** and **-1** in all other cases.

6. Use **Blue / Black Point Pen only** for writing particulars / marking responses on the Answer Sheet. **Use of pencil is strictly prohibited.**
7. No candidate is allowed to carry any textual material, printed or written, bits of papers, mobile phone any electron device etc, except the Identity Card inside the examination hall.
8. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
9. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
10. **Do not fold of make any stray marks on the Answer Sheet**

Name of the Candidate (in Capital): _____

Admission Number: | | | | | | | | | |

Candidate's Signature:_____ **Invigilator's Signature: _____**

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22-012026_Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT_Jee-Main_GTM-17_Test Syllabus

MATHEMATICS : Total Syllabus

PHYSICS : Total Syllabus

CHEMISTRY : Total Syllabus

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MATHEMATICS **Max Marks: 100**

SECTION-I
(SINGLE CORRECT ANSWER TYPE)

This section contains **20 Multiple Choice Questions.** Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

1. A value of θ for which $\frac{2+3i\sin\theta}{1-2i\sin\theta}$ is purely imaginary, is:

1) $\sin^{-1}\left(\frac{1}{\sqrt{3}}\right)$ **2)** $\frac{\pi}{3}$ **3)** $\frac{\pi}{6}$ **4)** $\sin^{-1}\left(\frac{\sqrt{3}}{4}\right)$

2. If the equations $x^2+2x+3=0$ and $ax^2+bx+c=0,\ a,b,c\in R,$ have a common root then $a:b:c$ is

1) 1:2:3 **2)** 3:2:1 **3)** 1:3:2 **4)** 3:1:2

3. The number of values of $k,$ for which the system of equations:

$(k+1)x+8y=4k : kx+(k+3)y=3k-1$ has no solution, is

1) infinite **2)** 1 **3)** 2 **4)** 3

4. The number of integers greater than 6,000 that can be formed, using the digits 3,5,6,7 and 8, without repetition, is:

1) 216 **2)** 192 **3)** 120 **4)** 72

5. The sum of first 20 terms of the sequence 0.7, 0.77, 0.777, ……., is

1) $\frac{7}{81}\left(179-10^{-20}\right)$ **2)** $\frac{7}{9}\left(99-10^{-20}\right)$

3) $\frac{7}{81}\left(179+10^{-20}\right)$ **4)** $\frac{7}{9}\left(99+10^{-20}\right)$

6. $\lim\limits_{x\to 0}\frac{\sin\left(\pi\cos^2 x\right)}{x^2}$ is equal to

1) $\frac{\pi}{2}$ **2)** 1 **3)** $-\pi$ **4)** π

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7. The area bounded by two parabolas $y^2 = 4ax$ and $x^2 = 4by$ is_____

1) $\frac{8ab}{3}$ **2)** $\frac{16ab}{3}$ **3)** $\frac{4ab}{3}$ **4)** $\frac{ab}{3}$

8. If $x = -1$ and $x = 2$ re extreme points of $f(x) = \alpha \log|x| + \beta x^2 + x$ then

1) $\alpha = -6, \beta = \frac{1}{2}$ **2)** $\alpha = -6, \beta = -\frac{1}{2}$ **3)** $\alpha = 2, \beta = -\frac{1}{2}$ **4)** $\alpha = 2, \beta = \frac{1}{2}$

9. Statement I : The value of the integral $\int_{\pi/6}^{\pi/3} \frac{dx}{1+\sqrt{\tan x}}$ is equal to $\frac{\pi}{6}$

Statement II : $\int_a^b f(x)dx = \int_a^b f(a+b-x)dx$

1) Statement I is true; statement II is true; statement II is a correct explanation for statement I

2) Statement I is true; statement II is true; statement II is a not correct explanation for statement I

3) Statement I is true; statement II is false

4) Statement I is false; statement II is true

10. The integral $\int \frac{2x^{12} + 5x^9}{\left(x^5 + x^3 + 1\right)^3} dx$ is equal to

1) $\frac{-x^{10}}{2\left(x^5 + x^3 + 1\right)^2} + C$ **2)** $\frac{-x^5}{\left(x^5 + x^3 + 1\right)^2} + C$

3) $\frac{x^{10}}{2\left(x^5 + x^3 + 1\right)^2} + C$ **4)** $\frac{x^5}{2\left(x^5 + x^3 + 1\right)^2} + C$

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11. Let PS be the median of the triangle with vertices $P(2,2), Q(6,-1)$ and $R(7,3)$. The equation of the line passing through $(1,-1)$ and parallel to PS is

1) $4x-7y-11=0$ **2)** $2x+9y+7=0$ **3)** $4x+7y+3=0$ **4)** $2x-9y-11=0$

12. $\sin^{-1}\left(\sin\frac{2\pi}{3}\right)=$ _____

1) $\frac{\pi}{3}$ **2)** $\frac{2\pi}{3}$ **3)** $\frac{-\pi}{3}$ **4)** $\frac{4\pi}{3}$

13. The equation of the circle passing through the foci of the ellipse $\frac{x^2}{16}+\frac{y^2}{9}=1$, and having centre at $(0,3)$:

1) $x^2+y^2-6y-7=0$ **2)** $x^2+y^2-6y+7=0$

3) $x^2+y^2-6y-5=0$ **4)** $x^2+y^2-6y+5=0$

14. The angle between the lines whose direction cosines satisfy the equations $l+m+n=0$ and $l^2=m^2+n^2$ is

1) $\frac{\pi}{3}$ **2)** $\frac{\pi}{4}$ **3)** $\frac{\pi}{6}$ **4)** $\frac{\pi}{2}$

15. If the vectors $\overline{AB}=3\hat{i}+4\hat{k}$ and $\overline{AC}=5\hat{i}-2\hat{j}+4\hat{k}$ are the sides of a triangle ABC, then the length of the median through A is

1) $\sqrt{18}$ **2)** $\sqrt{72}$ **3)** $\sqrt{33}$ **4)** $\sqrt{45}$

16. If $(2+\sin x)\frac{dy}{dx}+(y+1)\cos x=0$ and $y(0)=1$, then $y\left(\frac{\pi}{2}\right)$ is equal to:

1) $\frac{4}{3}$ **2)** $\frac{1}{3}$ **3)** $-\frac{2}{3}$ **4)** $-\frac{1}{3}$

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17. Let $f_k(x)=\frac{1}{k}\left(\sin^k x+\cos^k x\right)$ where $x\in R$ and $k\geq 1$, Then $f_4(x)-f_6(x)$ equals:

1) $\frac{1}{6}$ **2)** $\frac{1}{3}$ **3)** $\frac{1}{4}$ **4)** $\frac{1}{12}$

18. $\left(1+\cos\frac{\pi}{8}\right)\left(1+\cos\frac{3\pi}{8}\right)\left(1+\cos\frac{5\pi}{8}\right)\left(1+\cos\frac{7\pi}{8}\right)=$

1) $\frac{1}{2}$ **2)** $\cos\frac{\pi}{8}$ **3)** $\frac{1}{8}$ **4)** $\frac{1+\sqrt{2}}{2\sqrt{2}}$

19. The equation $\sqrt{x+1}-\sqrt{x-1}=\sqrt{4x-1}$ has

1) no solution **2)** one solution **3)** two solutions **4)** more than 2 solutions

20. If the coefficients of x^3 and x^4 in the expansion of $\left(1+ax+bx^2\right)(1-2x)^{18}$ in powers of $'x'$ are both zero, then (a,b) is equal to:

1) $\left(16,\frac{251}{3}\right)$ **2)** $\left(14,\frac{251}{3}\right)$ **3)** $\left(14,\frac{272}{3}\right)$ **4)** $\left(16,\frac{272}{3}\right)$

SECTION-II
(NUMERICAL VALUE TYPE

This section contains **5 Numerical Value Type Questions.** The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11**).

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases.

21. Let A and B be two sets containing 2 elements and 4 elements respectively. The number of subsets of $A\times B$ having 3 or more elements is

22. The variance of first 50 even natural numbers is

23. The number of common tangents to the circles $x^2+y^2-4x-6y-12=0$ and $x^2+y^2+6x+18y+26=0$, is

24. If the function. $g(x)=\begin{cases}k\sqrt{x+1}, & 0\leq x\leq 3\\ mx+2, & 3<x\leq 5\end{cases}$ is differentiable, then the value of $k+m$ is

25. The shortest distance between the skew lines $\frac{x+3}{-4}=\frac{y-6}{3}=\frac{z}{2}$ and $\frac{x+2}{-4}=\frac{y}{1}=\frac{z-7}{1}$ is equal to

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PHYSICS **Max Marks: 100**

SECTION-I
(SINGLE CORRECT ANSWER TYPE)

This section contains **20 Multiple Choice Questions.** Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

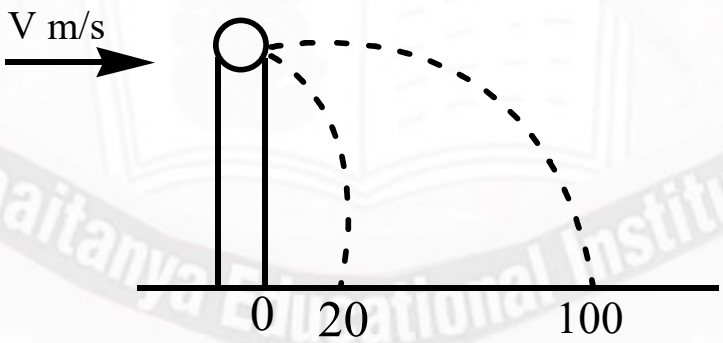
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26. A temperature difference can generate e.m.f. in some materials. Let S be the e.m.f. produced per unit temperature difference between the ends of a wire, σ is the electrical conductivity and k is the thermal conductivity of the material of the wire. Taking M,L,T, I and K as dimensions of mass, length, time, current and temperature, respectively, the dimensional formula of the quantity $Z = \frac{S^2\sigma}{k}$ is:

1) $\left[M^0L^0T^0I^0K^0\right]$ **2)** $\left[M^0L^0T^0I^0K^{-1}\right]$

3) $\left[M^1L^2T^{-2}I^{-1}K^{-1}\right]$ **4)** $\left[M^1L^2T^{-4}I^{-1}K^{-1}\right]$

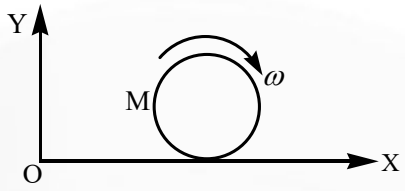
27. A ball of mass 0.2 kg rests on a vertical post of height 5 m. A bullet of mass 0.01 kg, traveling with a velocity V m/s in a horizontal direction, hits the centre of the ball. After the collision, the ball and bullet travel independently. The ball hits the ground at a distance of 20 m and the bullet at a distance of 100 m from the foot of the post. The velocity V of the bullet is



1) 250 m/s **2)** $250\sqrt{2} m/s$ **3)** $400\ m/s$ **4)** $500\ m/s$

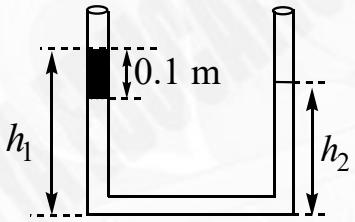
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28. A disc of mass M and radius R is rolling with angular speed ω on a horizontal plane as shown in figure. The magnitude of angular momentum of the disc about the origin O is



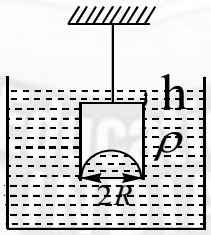
1) $(1/2)MR^2\omega$ **2)** $MR^2\omega$ **3)** $(3/2)MR^2\omega$ **4)** $2MR^2\omega$

29. An open-ended U-tube of uniform cross-sectional area contains water (density$10^3 kg\ m^{-3}$). Initially the water level stands at 0.29 m from the bottom in each arm. Kerosene oil (a water-immiscible liquid) of density 800 $kg\,m^{-3}$ is added to the left arm until its length is 0.1 m, as shown in the schematic figure below. The ratio $\left(\frac{h_1}{h_2}\right)$ of the heights of the liquid in the two arms is



1) $\frac{15}{14}$ **2)** $\frac{35}{33}$ **3)** $\frac{7}{6}$ **4)** $\frac{5}{4}$

30. A hemispherical portion of radius R is removed from the bottom of a cylinder of radius R. The volume of the remaining cylinder is V and its mass M. It is suspended by a string in a liquid of density ρ where it stays vertical. The upper surface of the cylinder is at a depth h below the liquid surface. The force on the bottom of the cylinder by the liquid is

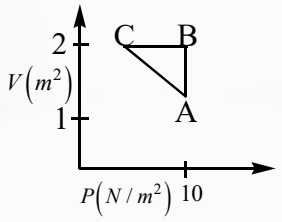


1) Mg **2)** $Mg - V\rho g$ **3)** $Mg + \pi R^2 h\rho g$ **4)** $\rho g\left(V + \pi R^2 h\right)$

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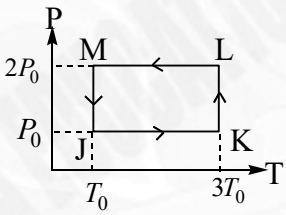

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31. An ideal gas is taken through the cycle $A \to B \to C \to A$, as shown in the figure. If the net heat supplied to the gas in the cycle is 5J, work done by the gas in the process $C \to A$ is



1) $-5J$ **2)** $-10J$ **3)** $-15J$ **4)** $-20J$

32. One mole of a monatomic ideal gas undergoes the cyclic process $J \to K \to L \to M \to J$, as shown in the $P-T$ diagram.



Match the quantities mentioned in List – I with their values in List – II and choose the correct option. [R is the gas constant.]

List – I		List – II	
P)	Work done in the complete cycle	1)	$RT_0 - 4RT_0 \ln 2$
Q)	Change in the internal energy of the gas in the process JK	2)	0
R)	Heat given to the gas in the process KL	3)	$3RT_0$
S)	Change in the internal energy of the gas in the process MJ	4)	$-2RT_0 \ln 2$
		5)	$-3RT_0 \ln 2$

1) $P \to 1; Q \to 3; R \to 5; S \to 4$ **2)** $P \to 4; Q \to 3; R \to 5; S \to 2$

3) $P \to 4; Q \to 1; R \to 2; S \to 2$ **4)** $P \to 2; Q \to 5; R \to 3; S \to 4$

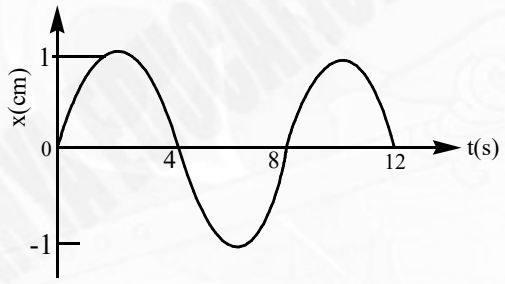
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33. Consider a spherical gaseous cloud of mass density $\rho(r)$ in a free space where r is the radical distance from its center. The gaseous cloud is made of particles of equal mass m moving in circular orbits about the common centre with the same kinetic energy K. The force acting on the particles is their mutual gravitational force. If $\rho(r)$ is constant in time. The particle number density $n(r)=\rho(r)/m$ is

[G is universal gravitational constant]

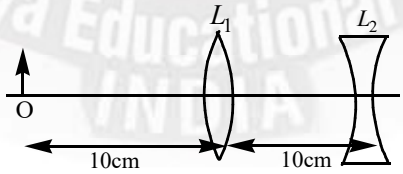
1) $\frac{3K}{\pi r^2 m^2 G}$ **2)** $\frac{K}{2\pi r^2 m^2 G}$ **3)** $\frac{K}{\pi r^2 m^2 G}$ **4)** $\frac{K}{6\pi r^2 m^2 G}$

34. The $x-t$ graph of a particle undergoing simple harmonic motion is shown below. The acceleration of the particle at $t=4/3\ s$ is



1) $\frac{\sqrt{3}}{32}\pi^2 cm/s^2$ **2)** $\frac{-\pi^2}{32}cm/s^2$ **3)** $\frac{\pi^2}{32}cm/s^2$ **4)** $-\frac{\sqrt{3}}{32}\pi^2 cm/s^2$

35. An extended object is placed at point O, 10 cm in front of a convex lens L_1 and a concave lens L_2 is placed 10 cm behind it, as shown in the figure. The radii of curvature of all the curved surfaces in both the lenses are 20 cm. The refractive index of both the lenses is 1.5. The total magnification of this lens system is

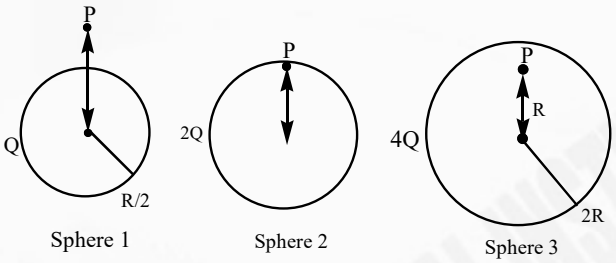


1) 0.4 **2)** 0.8 **3)** 1.3 **4)** 1.6

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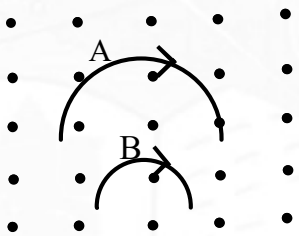

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36. Charges $Q, 2Q$ and $4Q$ are uniformly distributed in three dielectric solid spheres 1, 2 and 3 of radii $R/2$, R and $2R$ respectively, as shown in figure. If magnitude of the electric fields at point P at a distance R from the centre of sphere 1, 2 and 3 are E_1, E_2 and E_3 respectively, then



1) $E_1 > E_2 > E_3$ **2)** $E_3 > E_1 > E_2$ **3)** $E_2 > E_1 > E_3$ **4)** $E_3 > E_2 > E_1$

37. Two particles A and B of masses m_A and m_B respectively and having the same charge are moving in a plane. A uniform magnetic field exists perpendicular to this plane. The speeds of the particles are v_A and v_B respectively and the trajectories are as shown in the figure. Then

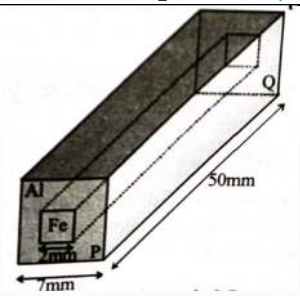


1) $m_A v_A < m_B v_B$ **2)** $m_A v_A > m_B v_B$

3) $m_A < m_B$ and $v_A < v_B$ **4)** $m_A = m_B$ and $v_A = v_B$

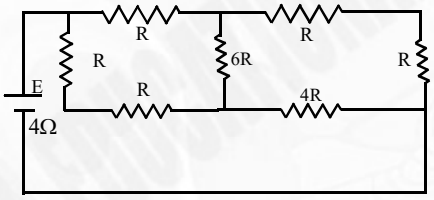
38. In an aluminium (Al) bar of square cross section, a square hole is drilled and is filled with iron (Fe) as shown in the figure. The electrical resistivities of Al and Fe are $27 \times 10^{-8} \Omega m$ and $1.0 \times 10^{-7} \Omega m$, respectively. The electrical resistance between the two faces P and Q of the composite bar is

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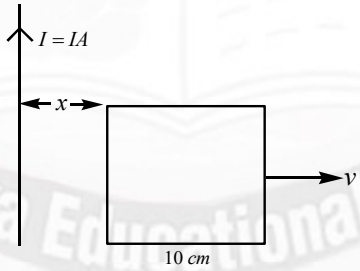
1) $\frac{2475}{64}\mu\Omega$ **2)** $\frac{1875}{64}\mu\Omega$ **3)** $\frac{1875}{49}\mu\Omega$ **4)** $\frac{2475}{132}\mu\Omega$

39. A battery of internal resistance 4Ω is connected to the network of resistances as shown. In order that the maximum power can be delivered to the network, the value of R in Ω should be



1) $\frac{4}{9}$ **2)** 2 **3)** $\frac{8}{3}$ **4)** 18

40. A square frame of side 10 cm and a long straight wire carrying current 1 A are in the plate of the paper. Starting from close to the wire, the frame moves towards the right with a constant speed of 10 ms^{-1} (see figure).

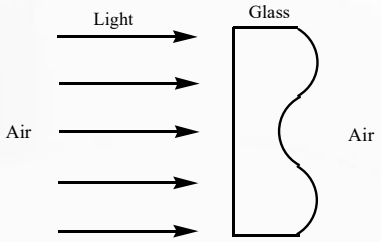


The e.m.f induced at the time the left arm of the frame is at $x = 10\ cm$ from the wire is:

1) $2\mu V$ **2)** $1\mu V$ **3)** $0.75\mu V$ **4)** $0.5\mu V$

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41. A parallel beam of light strikes a piece of transparent glass having cross section as shown in the figure below. Correct shape of the emergent wavefront will be (figures are schematic and not drawn to scale)



1)  **2)**  **3)**  **4)** 

42. For Young's double slit experiment, two statements are given below:

Statement I: If screen is moved away from the plane of slits, angular separation of the fringes remains constant.

Statement II: If the monochromatic source is replaced by another monochromatic source of larger wavelength, the angular separation of fringes decreases.

In the light of the above statements choose the correct answer from the options given below:

1) Both Statement I and Statement II are true

2) Both Statement I and Statement II are false

3) Statement I is true but Statement II is false

4) Statement I is false but Statement II is true

43. Assertion (A): The dominant mechanism for motion of charge carriers in forward and reverse biased silicon p-n junction are drift in both forward and reverse bias.

Reason (R): In reverse biasing, no current flow through the junction

1) Both (A) and (R) are true and R is the correct explanation of A

2) Both (A) and (R) are true and R is not the correct explanation of A

3) (A) is true, (R) is false

4) Both (A) and (R) are false

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44. Assertion (A): The fusion process is achieved by raising the temperature of the system

Reason (R): The fusion particles have enough K.E to overcome the coulomb repulsive behaviour.

1) Both (A) and (R) are true and R is the correct explanation of A

2) Both (A) and (R) are true and R is not the correct explanation of A

3) (A) is true, (R) is false

4) (A) false (R) is true

45. Statement I : In hydrogen atom, the frequency of radiation emitted when an electron jumps from lower energy orbit (E_1) to higher energy orbit (E_2), is given as $hf = E_1 - E_2$.

Statement II: The jumping of electron from higher energy orbit (E_2) to lower energy orbit (E_1) is associated with frequency of radiation given as $f = (E_2 - E_1)/h$

The condition is Bohr's frequency condition.

In the light of the above statements, choose the correct answer from the options given below:

1) Both Statement I and Statement II are true

2) Both Statement I and Statement II are false

3) Statement I is true but Statement II is false

4) Statement I is false but Statement II is true

SECTION-II
(NUMERICAL VALUE TYPE

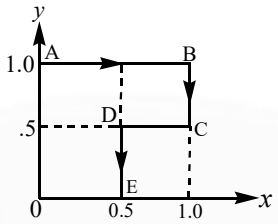
This section contains **5 Numerical Value Type Questions.** The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases

46. A particle is moved along a path $AB \to BC \to CD \to DE$ as shown in figure, in presence of a forced $\vec{F} = (\alpha y\hat{i} + 2\alpha x\hat{j})N$, where x and y are in meter and $\alpha = -1Nm^{-1}$. The work done on the particle by this force $\vec{F}$ will be_____ Joule.

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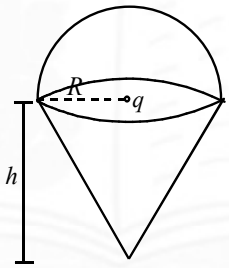

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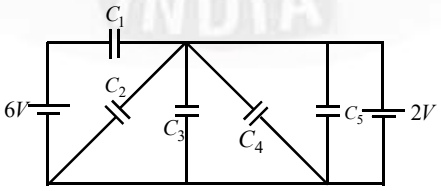
47. A particle of mass 1 kg is subjected to a force which depends on the position as $\vec{F} = -k\left(x\hat{i} + y\hat{j}\right) kg\ ms^{-2}$ with $k = 1\ kg\ s^{-2}$. At time $t = 0$, the particle's position $\vec{r} = \left(\frac{1}{\sqrt{2}}\hat{i} + \sqrt{2}\hat{j}\right) m$ and its velocity $\vec{v} = \left(-\sqrt{2}\hat{i} + \sqrt{2}\hat{j} + \frac{2}{\pi}\hat{k}\right) ms^{-1}$. Let v_x and v_y denote the x and the y components of the particle's velocity, respectively. Ignore gravity. When $z = 0.5m$, the value of $\left(xv_y - yv_x\right)$ is_____ m^2s^{-1}.

48. Image of an object approaching a convex mirror of radius of curvature 20 m along its optical axis is observed to move from $\frac{25}{3}m$ *to* $\frac{50}{7}m$ in 30 seconds. What is the speed of the object in km per hour?

49. A charge q is surrounded by a closed surface consisting of an inverted cone of height h and base radius R, and a hemisphere of radius R as shown in the figure. The electric flux through the conical surface is $\frac{nq}{6\in_0}$ (in SI units). The value of n is_____



50. In the following circuit $C_1 = 12\mu F, C_2 = C_3 = 4\mu F$ and $C_4 = C_5 = 2\mu F$. The charge stored in C_3 is_____ μC.



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CHEMISTRY

Max Marks: 100

SECTION-I
(SINGLE CORRECT ANSWER TYPE)

This section contains **20 Multiple Choice Questions.** Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

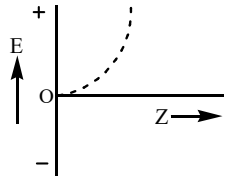
51. $CaCO_3(s)+2HCl(aq)\rightarrow CaCl_2(aq)+CO_2(g)+H_2O(1)$ consider the above reaction, what mass of $CaCl_2$ will be formed if 250 mL of 0.76 M HCl reacts with 1000 g of $CaCO_3$?

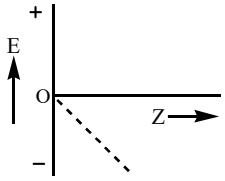
(Given : Molar mass of Ca, C, O, H and Cl are 40,12,16,1 and 35.5 $g\ mol^{-1}$, respectively)

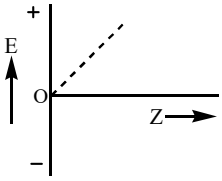
1) 3.908 g **2)** 2.636 g **3)** 10.545 g **4)** 5.272 g

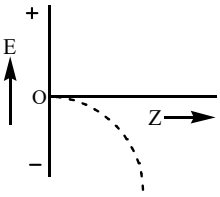
52. For hydrogen like species, which of the following graphs provides the most appropriate representation of E vs Z plot for a constant n?

[E: Energy of the stationary state, Z: atomic number, n = principal quantum number]

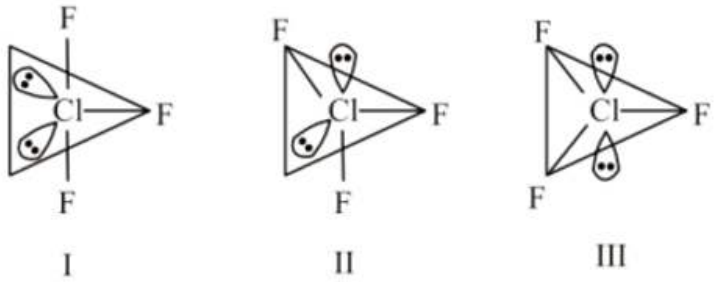
1) 

2) 

3) 

4) 

53. Statement (I): For $\ddot{C}l\ddot{F}_3$, all three possible structures may be drawn as follows.



Statement II: Structure III is most stable, as the orbitals having the lone pairs are axial, where the $lp-bp$ repulsion is minimum.

In the light of the above statements, choose the most appropriate answer from the options given below:

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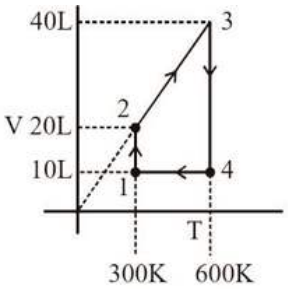
1) Statement I is incorrect but statement II is correct

2) Statement I is correct but statement II is incorrect

3) Both statement I and statement II are correct

4) Both statement I and statement II are incorrect

54. What is the net work done when 1 mole of monatomic ideal gas undergoes in a process described by 1, 2, 3, 4 in given V-T graph. **(Use: R = 2cal/mole K, ln 2=0.7)**



1) – 600 cal **2)** – 660 cal **3)** + 660 cal **4)** + 600 cal

55. Statement I: Addition of ethylene glycol (non-volatile) to water lowers the freezing point of water hence used as antifreeze.

Statement II: Addition of any nonvolatile substance to water lowers its freezing point of water.

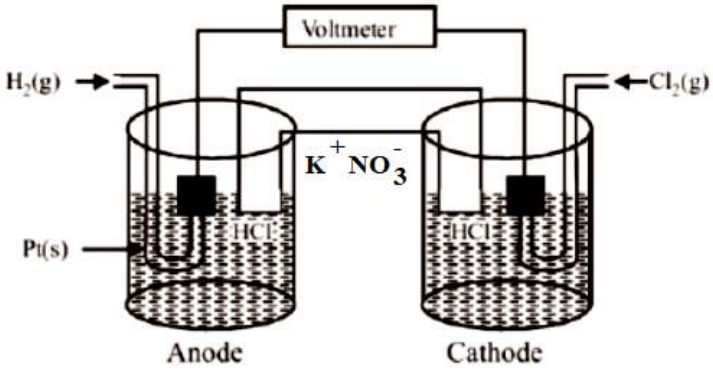
1) Statement I is true, statement II is true and statement II is correct explanation for statement I

2) Statement I is true, statement II is true and statement II is NOT the correct explanation for statement I

3) Statement I is true, statement II is false

4) Statement I is false, statement II is true

56. Consider the following Galvanic cell.



By what value the cell voltage change when concentration of ions in anodic and cathodic compartments both increased by factor of 10 at 298 K

1) +0.0590 **2)** – 0.0590 **3)** – 0.1180 **4)** 0

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57. The kinetic data for the given reaction $A(g)+2B(g)\rightarrow C(g)$ is provided in the following table for three experiments at 300 K.

Ex. No.	$[A/M]$	$[B/M]$	[Initial rate $(M\ \sec^{-1})$]
1	0.01	0.01	6.930×10^{-6}
2	0.02	0.01	1.386×10^{-5}
3	0.02	0.02	1.386×10^{-5}

In another experiment starting with initial concentration of 0.5 and 1 M respectively for A and B at 300 K. Find the rate of reaction after 50 minutes from start of experiment (in m/sec)?

1) 6.93×10^{-4} **2)** 0.25×10^{-7} **3)** 4.33×10^{-5} **4)** 3.46×10^{-9}

58. Match the List – I with List – II

List – I (Family)		List – II (Symbol of Element)	
A.	Pnictogen (group 15)	**I.**	*Ts*
B.	Chalcogen	**II.**	*Og*
C.	Halogen	**III.**	*Lv*
D.	Noble gas	**IV.**	*Mc*

Choose the correct answer from the options given below:

1) $A-IV;B-I;C-II;D-III$ **2)** $A-IV;B-III;C-I;D-II$

3) $A-III;B-I;C-IV;D-II$ **4)** $A-II;B-III;C-IV;D-I$

59. Assertion (A) : The stability order of +1 oxidation state of Ga, In and Tl is $Ga<In<Tl$.
Reason (R): The inert pair effect stabilizes the lower oxidation state down the group.
In the light of the above statements, choose the correct answer from the options given below:

1) Both A and R are true and R is the correct explanation of A

2) A is true but R is false

3) Both A and R are true but R is NOT the correct explanation of A

4) A is false but R is true

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60. Statement I: H_2Se is more acidic than H_2Te

Statement II: H_2Se has higher bond enthalpy for dissociation than H_2Te.

In the light of the above statements, choose the correct answer from the options given below.

1) Both statement I and statement II are false

2) Both statement I and statement II are true

3) Statement I is true but statement II is false

4) Statement I is false but statement II is true

61. Which of the following statements are not correct when a mixture of $NaCl$ and $K_2Cr_2O_7$ is gently warmed with conc. H_2SO_4?

1) Deep red vapours are liberated

2) Deep red vapours dissolve in $NaOH$ forming a yellow solution

3) Greenish yellow gas is liberated

4) Deep red vapours dissolve in water forming yellow solution

62. Match each coordination compound in List – I with co-ordination number of central metal/ion from List – II and select the correct answer using the code given below the lists.

List – I		List – II	
P.	$\left[Co(en)_3\right]^{3\oplus}$	1)	6
Q.	$\left[Ca(EDTA)\right]^{2\Theta}$	2)	4
R.	$\left[Ni(CO)_4\right]$	3)	2
S.	$\left[Ag(NH_3)_2\right]Cl$	4)	5

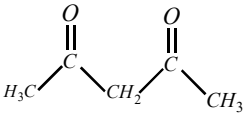
1) $P \to 1; Q \to 3; R \to 2; S \to 3$ **2)** $P \to 1; Q \to 1; R \to 2; S \to 3$

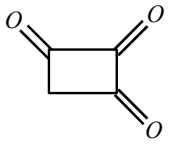
3) $P \to 2; Q \to 1; R \to 2; S \to 2$ **4)** $P \to 2; Q \to 2; R \to 3; S \to 2$

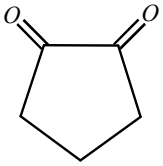
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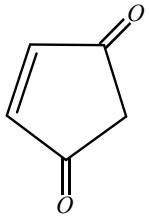

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63. Which of the following compounds has negligible enol?

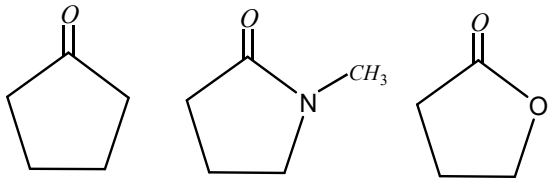
1) 

2) 

3) 

4) 

64. Arrange the following compounds in order of their reactivity toward $LiAlH_4$.



i ii iii

1) $i < ii < iii$ **2)** $i < iii < ii$ **3)** $ii < i < iii$ **4)** $ii < iii < i$

65. The percentage of sulphur in an organic compound whose 0.32 g produces 0.233 g of $BaSO_4$ [At.wt. Ba=137, S=32] is

1) 1.0 **2)** 10.0 **3)** 23.5 **4)** 32.1

66. The Prussian blue colour obtained during the test of nitrogen by Lassaigne's test is due to the formation of:

1) $Fe\left[Fe(CN)_6\right]_3$ **2)** $Na_3\left[Fe(CN)_6\right]$

3) $Fe(CN)_3$ **4)** $Fe_4\left[Fe(CN)_6\right]_3$

67. HBr reacts with $CH_2 = CH - OCH_3$ at room temperature to give

1) $BrCH_2CHO$ and CH_3OH **2)** $BrCH_2 - CH_2 - OCH_3$

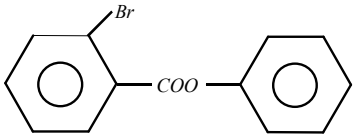
3) $H_3C - CHBr - OCH_3$ **4)** CH_3CHO and CH_3Br

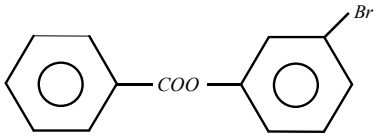
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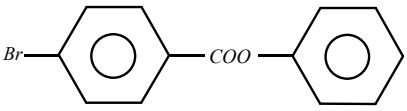


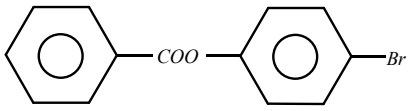
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68. The major product expected from the mono-bromination of phenyl benzoate is:

1) 

2) 

3) 

4) 

69. At iso-electric point:

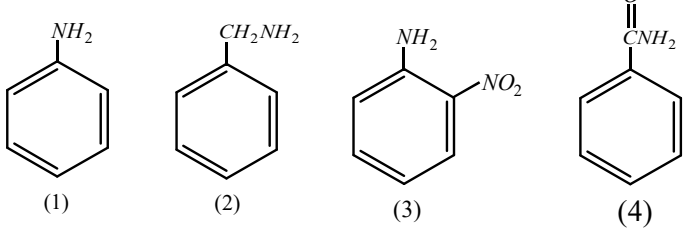
1) Concentration of cation is equal to concentration of anion

2) Net charge is zero

3) Maximum conc. of di-pole ion (Zwitter ion) will be present

4) All of the above

70. Rank the following compounds in order of increasing basic strength. (weakest → strongest):



1) $4<2<1<3$ **2)** $4<3<1<2$ **3)** $4<1<3<2$ **4)** $2<1<3<4$

SECTION-II
(NUMERICAL VALUE TYPE

This section contains **5 Numerical Value Type Questions.** The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases

71. 1 mole of ideal gas is allowed to expand reversibly and adiabatically from a temperature of 27^0C. The work done is $3\ kJ\ mol^{-1}$. The final temperature of the gas is_____ K (Nearest integer). Given $C_v = 20\ J\ mol^{-1}K^{-1}$.

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72. Consider the following reaction approaching equilibrium at 27^0C and 1 atm pressure

$$A + B \underset{K_r=10^2}{\overset{K_f=10^3}{\rightleftharpoons}} C + D$$

The standard Gibb's energy change $\left(\Delta_r G^0\right)$ at 27^0C is $(-)$_____ $kJ\ mol^{-1}$

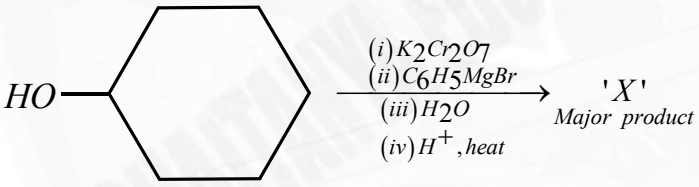
(Nearest integer).

(Given : $R = 8.3\ JK^{-1}mol^{-1}$ and $\ln 10 = 2.3$)

73. 150 g of acetic acid was contaminated with 10.2 g ascorbic acid $\left(C_6H_8O_6\right)$ to lower down its freezing point by $\left(x \times 10^{-1}\right)^0 C$. The value of x is_____. (Nearest integer)

[Given $K_f = 3.9\,K\ kg\ mol^{-1}$; Molar mass of ascorbic acid $= 176\,g\ mol^{-1}$]

74. In the given reaction



The number of sp^2 hybridized carbon(s) in compound 'X' is_____

75. The number of tri-peptides formed by three different amino acids using each amino acid once is_____.

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